

IM3180 Design and Innovation Project (AY2025/26 Semester 1)

Individual Report

Name: _____ Hiew Wei Feng_____

Group: _____ IE08_____

Project Title: _____ DeciWise_____

Contributions to the Project and Report (1-2 pages)

Project

1. Research on market available products, competitors and existing solutions for leadership training in corporate settings.
2. Research on crisis management and PR trained response .
3. Jira Updating and Project Management
4. Obtaining GAI Image Generation API (Gemini 2.5 Flash Nanobanana)
5. Implementation and draft code for Image Generation with Gemini enabled Python Library (google-generativeai)
6. Testing and integration of Image-Gen function into main flowchart function, extensive discussion on JSON format for file storing and reading.
7. Integrating non-user called JSON file reading directly to improve efficiency of Image Generation Function to all nodes.
8. Project Pitching and presentation

Report (indicate pages and appendixes written by you for the Group Report)

1. Section 1, (1.1-1.5)
2. Section 3, (3.1-3.3)
3. Section 4, (4.1.4, 4.2)
4. Appendix C (Design Diagrams)

Other Deliverables like Project Charter (indicate pages and appendixes), Poster, Presentation Slides (indicate slide numbers) and Video, if applicable

1. Project Charter – Project Description, Budget Summary
2. Video Production – Filming and props preparation
3. Presentation Slides – slide 2 – slide 9

Reflection on Learning Outcome Attainment

Reflect on your experience during your project and the achievements you have relating to at least two of the points below:

- (a) Engineering knowledge
- (b) Problem Analysis
- (c) Investigation
- (d) Design/development of Solutions
- (e) Modern Tool Usage
- (f) The Engineer and the World
- (g) Ethics
- (h) Individual and Collaborative Team Work
- (i) Communication
- (j) Project Management and Finance
- (k) Lifelong Learning

Point 1: b,c State the area

This project started from our intention to create a platform to solve a market available challenge/issue, after doing some research on current issues, and further discussions with our team and prof Andy, we have decided and been suggested to create a platform that allows potential leaders/corporate staff to have an alternative towards traditional leadership training. After extensive research on market shifts and global intention and initiatives on leadership training for leaders, we have slimmed down to focus on an area, that is strategic decision making, as it emerges as one of the critical skills that forms a successful, sustainable organization. These significant reasons has been a strong foundation, and a valid problem statement for the development of our project.

Throughout the project, I have been also partaking in research that includes psychological and crisis management in PR training, alongside with market-available products that attempts to address an alternative to traditional leadership training methods. Through these experiences, I have sharpened my skills and ability to capture and identify relevant key points, along with validating reputable sources for our project. These sources proved further that our product would be relevant towards the intended audiences, and allowed us to focus on combining multiple different functions that the market available products do not offer.

Point 2: h State the area Individual and collaborative teamwork

From my past experiences, particularly IDP, Intro to Design and Project, which we worked in teams of 2. This project, as compared to the project in IDP, was difficult in many different aspects, which involved many parties working together on different sectors of project development, for example, research, analysis, project management, coding, testing, and documentation. These aspects further proved that collaboration remains one of the key factors in developing such a large project. Personally, I was involved in all the different sectors, from research to documentation, and understood that I would prefer more of a practical role, as compared to a technical role.

In terms of coding, I have been developing the Image generation part leveraging Gemini's 2.5 Flash Image (Nanobanana), this also allowed me to understand the usage of API keys in project development, which I have personally also been interested in. After developing this component, and extensive discussions on combining this function with Sriya's flowchart generation using GPT 4o-mini. With further tweaks and testing, we managed to launch it as one whole component, with multiple

functions under the same file, and implementing the flask app which allows data transfer from front to backend, allows the whole AI integration of the project to come to life.

Aside from these, I have also learnt and understood the benefits of using Github for version control and collaboration. As previously, I did not have to manage such a big project, we just worked on our individual parts, and combined it using a pendrive to compile into a single file. This method has proved to be confusing and requires additional debugging (Due to inconsistent code format and variables) as compared to using GitHub for instantaneous debugging.

Acknowledging/Declaring the Use of GAI

Please refer to NTU's Current Policy & Guidelines on the Use of Generative AI available in NTUlearn home page and the link:

[https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-\(GAI\)-Tools-on-Education.aspx](https://entuedu.sharepoint.com/sites/Student/dept/ctlp/SitePages/Exploring-the-Impact-of-Generative-Artificial-Intelligence-(GAI)-Tools-on-Education.aspx)

- Complete the following declaration if applicable.
- Create a Paper Trail to document the input prompt, output obtained, and how you have used it

I _____ Hiew Wei Feng _____ (student name),
_____ Hiew0012 _____@e.ntu.edu.sg (NTU email) honestly and sincerely make the following declaration in relation to the following course submission:

1. Name of course: Design & Innovation Project
2. Course Code: IM3180
3. Instructor: Prof Andy Khong
4. Title of Assignment/Project Submission: DIP Individual Report

In relation to the foregoing I hereby declare that, fully and properly in accordance with the Assignment/Project Instructions I have (check where appropriate):

- i. Used GAI as permitted to assist in generating key ideas only. ☐
- ii. Used GAI as permitted to assist in generating a first text only. ☐
- And/or
- iii. Used GAI to refine syntax and grammar for correct language submission only. ☐
- Or
- iv. As it is not permitted: Not used GAI assistance in any way in the development or generation of this assignment or project. ☒

I also declare that I have :

- a. Fully and honestly submitted the digital paper trail required under the assignment/project instructions; and that
- b. Wherever GAI assistance has been employed in the submission in word or paraphrase or inclusion of a significant idea or fact suggested by the GAI assistant, I have acknowledged this by a footnote; and that,
- c. Apart from the foregoing notices, the submission is wholly my own work.



Student Name & Signature

14/11/2025
Date